

01 – Organisation

Several information about the course

(based on Andreas Axelsson's version of the course
with nice and efficient help from Alexander Göransson)

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OUTLINE

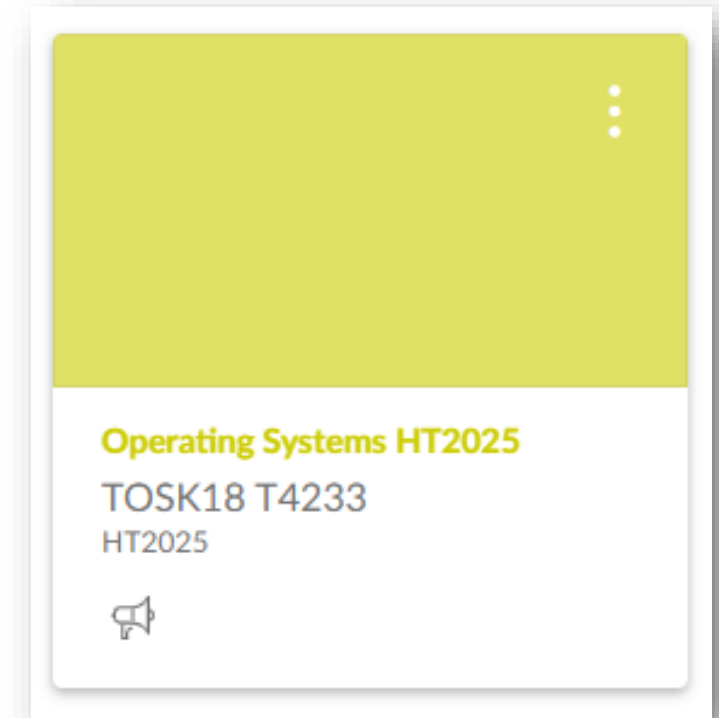
- 1) Organisation
- 2) Moral Contract
- 3) Syllabus
- 4) Literature
- 5) Schedule
- 6) Course Information
- 7) Groups
- 8) Grading
- 9) General information

1) ORGANISATION

- **TOSK18 - Operating Systems**
- The course is given in **ENGLISH** (för att min svenska är inte bra nu)
- The course will last for 7 weeks + 1 week of examination
- Every week, you will have:
 - 2 sessions (2 x 45 min. each) lectures/seminars/workshops or less...
 - 1 lab (03:45 each)
- I hope to have at least one seminar on RTOS by Husqvarna Engineers
- 2 Lab Assistants: ?, Jérôme Landré.
- When discussing with teachers, please follow the code of conduct at JU: respect persons and stay polite and calm even if you are unhappy (especially if you are unhappy)

1) ORGANISATION

- **Canvas** is used during the course
- **Announcements** will be sent in case of changes in the course!
- For now:
 - Study guide
 - Syllabus
- Documents will be added on-the-fly



2) MORAL CONTRACT

- A course is a moral contract between a teacher and a student (many students):
 - Teacher must:
 - Give useful information to the students, knowledge and methodology
 - Guide their learning to ensure the ILOs
 - Assess the students understood the course
 - Students must:
 - Do the job
 - Attend the lectures and labs
 - Succeed
- "The teacher is a facilitator", Björn Praestgaard-Larsen
- "The student is responsible for his/her/their learning"

2) MORAL CONTRACT

- In this course, you **CAN** use ChatGPT and other chatbots to **HELP** you.
- **BUT:**
 - 1) you **MUST check** the validity of the answer on your own
 - 2) you **MUST understand** the given answer!
- Chatbots are very good tools to help the students but they can make mistakes (hallucinations)!

3) SYLLABUS

- Introduction to Operating Systems (OS):
 - To give basic knowledge of operating systems for real time and embedded systems and how these can be configured and used by their programming interfaces.
- The course covers the following topics:
 - Introduction to real time systems
 - Structure of operating systems, especially those with real time capabilities
 - Concurrent programming
 - Scheduling of real time tasks
 - Use of operating systems

3) SYLLABUS (ILOS)

- After the course, the student will:
 - Demonstrate an ability to use an operating system when developing in an embedded system.
 - Have knowledge about and be able to explain how operating systems are constructed and works , especially when it comes to interrupts , processes , threads , and scheduling
 - Have knowledge about what characterizes different kinds of real time systems
 - Have knowledge about and be able to explain the interaction between hardware and operating system
 - Have knowledge about and be able to describe some operating systems for embedded systems
 - Have knowledge about how interrupts , jitter , and drift etc affects a real time systems and how the effects can be minimized
 - Have knowledge about how Linux operating systems are built especially regarding architecture , configuration , process handling, and file systems etc.
 - Demonstrate the ability to use different methods to decide if scheduling is feasible

3) SYLLABUS

- Type of instruction:
 - Teaching consists of lectures, seminars, workshops, live coding sessions, and laboratory exercises
- Prerequisites
 - Completed courses in “Introduction to programming”, 7.5 credits
- Examination and grading
 - The course is graded Fail (U), 3, 4 or 5
- Registration of examination:

Name of Test	Value	Grading
Examination ⁽¹⁾	4 credits	U/3/4/5
Laboratory work	3.5 credits	U/G

⁽¹⁾ Determines the final grade of the course, which is issued only when all course units have been passed

4) LITERATURE

- Recommended (not mandatory) books:
 - "Modern Operating Systems", Andrew S. Tanenbaum , Herbert Bos
 - ISBN: 9781292061429
 - "Hard Real Time Computing Systems", Giorgio C. Buttazzo
 - ISBN: 9781461430193
- Other
 - Several online resources will be linked in Canvas

5) SCHEDULE

- Proposition for the schedule (can change a little during the course):

- **Lectures**

- Week 44 – Introduction & C
 - Week 45 – C
 - Week 46 – Bases & Processes and Threads
 - Week 47 – Linux commands
 - Week 49 – Scheduling & Synchronisation
 - Week 50 - FreeRTOS
 - Week 51 - Written exam

- **Labs**

- Lab0
 - Lab1
 - Lab2
 - Lab3
 - Lab4
 - Late Lab presentations

6) COURSE INFORMATION

- Lectures (1/2 per week)
 - Before each lecture, read the previous lecture to **memorize information**
 - During the lectures, **useful information** will be given:
 - On the slides: well presented little engagement from students
 - On the whiteboard: not well presented higher engagement from students

COME TO THE LECTURES!

6) COURSE INFORMATION

- Laboratory work
 - Group of 2 persons
 - **Read labs and prepare in advance!**
 - You **MUST** submit all your labs codes on Canvas
 - You **MUST** present several labs to the teaching assistants (be **two** to present)



6) COURSE INFORMATION

- Laboratory work
 - Group of 2 persons
 - **Read labs and prepare** in **advance**!
 - You **MUST** submit all your labs codes on Canvas
 - You **MUST** present several labs to the teaching assistants (be **two** to present)

COME TO THE LABS!



6) COURSE INFORMATION

- Laboratory work
 - Group of 2 persons
 - **Read labs and prepare in advance!**
 - You **MUST** submit all your labs codes on Canvas
 - You **MUST** present several labs to the teaching assistants (be **two** to present)



COME TO THE LABS!
AS A TEAM OF 2 PERSONS!

6) COURSE INFORMATION

- Software:
 - Oracle VirtualBox



- (Material)
 - Raspberry Pi 400 (1 per lab group of two persons)



7) GROUPS

- Groups of two students:

One from Digital and Mobile Platform (DMP) **with** One from Digital Information Security (DIS)/exchange programs/other (when possible of course)

- Rules:

- Be polite!
- Communicate!
- Agree with hours and respect them!
- Work as a team (50%/50%)
- Try to support your teammate as he/she/they support(s) you

8) GRADING

- Written exam:
 - Fail (U), 3, 4 or 5 --- 3 being the lowest to pass the course and 5 being the highest grade
 - Inspira Digital Exam:
 - 100 points questionnaire about Operating Systems, Linux, and RTOS
 - Questions will be related to the lectures, the programming codes, the labs
 - Grade 3: points ≥ 50
 - Grade 4: points ≥ 75
 - Grade 5: points ≥ 90
- Labs presentation:
 - Fail (U) / Pass (G)
- Overall grade: you need a Pass for Labs **and** at least grade 3 in the written exam to pass the course

9) GENERAL INFORMATION

- JU student webben: <https://ju.se/student/en.html>
- Examination: <https://ju.se/student/en/studies/examination.html>
- Academic Year: <https://ju.se/student/en/studies/academic-calendar/school-of-engineering.html>
- Student life: <https://ju.se/student/en/studies/student-life-and-the-student-union.html>
- Equal opportunities: <https://ju.se/student/en/studies/equal-opportunities-at-ju.html>
- Code of Conduct: <https://ju.se/student/en/studies/equal-opportunities-at-ju/code-of-conduct.html>

9) GENERAL INFORMATION

- In case of **Emergency**: <https://ju.se/student/en/campus/in-case-of-emergencies.html>
- Service center: <https://ju.se/student/en/service/service-center.html>
- Student Health Care: <https://ju.se/student/en/service/student-health-care.html>
- Support for students with disabilities: <https://ju.se/student/en/service/support-for-students-with-disabilities.html>
- Academic resource center: <https://ju.se/student/en/service/academic-resource-center-studieverkstaden.html>

ANY QUESTIONS?